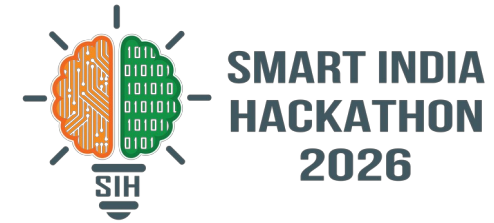


SMART INDIA HACKATHON 2026



TwinGuard Aero

Problem Statement ID - SIH26054

Problem Statement Title

AI-Enabled Real-Time Digital Twin System for Health Monitoring, Fault Prediction and Mission Reliability Enhancement of Aero Piston Engines used in MALE UAVs.

Theme - Robotics and Drones

PS Category - Software

Team ID - CONFIRM FROM PORTAL

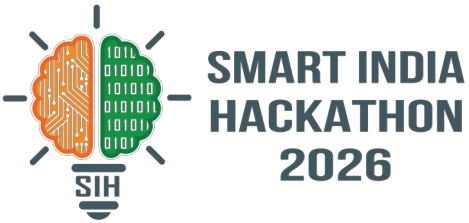
Team Name - MINDMESH

Synthetic prototype | Engine-specific validation planned





TwinGuard Aero



Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

A synchronized aero-piston twin combines a healthy reference, residual trends, Sensor Trust and explainable condition assessment.

How it addresses the problem

Inspect the evidence behind a warning; compare the same mission before and after degradation; retain observations for review.

Innovation and uniqueness of the solution

Connect measurement trust, subsystem evidence and profile consequences in one traceable operator workflow.

TRUST

Freshness + signal consistency

CONDITION

Residuals + temporal evidence

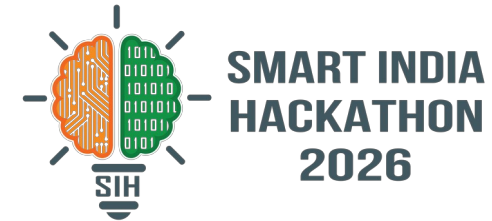
MISSION

State-dependent profile
comparison

Synthetic evidence; human review



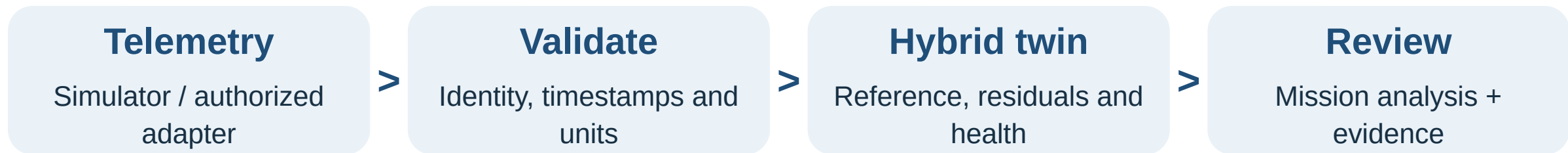
TECHNICAL APPROACH



Technologies to be used (e.g. programming languages, frameworks, hardware)

React / TypeScript + Three.js; Python / FastAPI + WebSockets; SQLite locally; Docker / PostgreSQL path.
Optional synthetic Isolation Forest and XGBoost models.

Methodology and process for implementation (Flow Charts/Images/ working prototype)



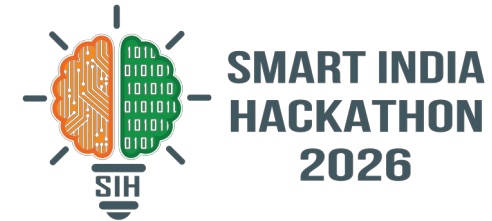
HOSTED: browser synthetic demonstration | LOCAL: authenticated backend + persistence

Evaluation Center: baseline capture → progressive fault → comparison → source-labeled JSON export.

Freshness and quality gate decisions. Insufficient live evidence produces DATA HOLD.



FEASIBILITY AND VIABILITY



Analysis of the feasibility of the idea

Implemented software demonstrator: telemetry contract, explainable analytics, mission comparison, replay and evidence export. Modular replacement of the generic reference supports later calibration.

Potential challenges and risks

No authorized real-engine labels yet. Physics and synthetic training share assumptions. RUL bands and mission scores are uncalibrated. Current runtime serves one configured engine.

Strategies for overcoming these challenges

Read-only rig integration → verified channels and independent labels → engine-map calibration → held-out threshold/hybrid comparison → hardware-in-the-loop and security review.

Planned measures: false alarms/hour, fault recall, warning lead time, RUL coverage and hardware latency.

Potential impact on the target audience

UAV operators

See data readiness and the evidence behind a changing engine condition.

Propulsion engineers

Inspect operating-context residuals and compare modelled profiles.

Maintenance teams

Review recorded events, probable conditions and engineering guidance.

Benefits of the solution (social, economic, environmental, etc.)

Reliability research

Earlier engineering review and better use of condition history.

Engineering efficiency

A repeatable scenario and reusable evidence trail for diagnosis.

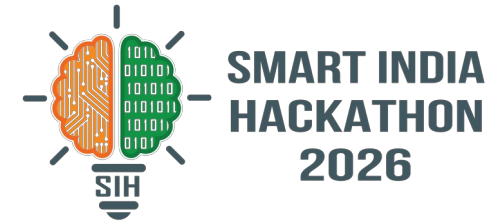
Resource stewardship

Potential for condition-led maintenance; savings require field measurement.

Intended benefits, not measured fleet outcomes. No claimed percentage reduction in failures, cost or fuel.



RESEARCH AND REFERENCES



Details / Links of the reference and research work

Official SIH26054 problem statement

sih.gov.in/sih2026PS

Scope and deliverables for the selected DRDO challenge.

NIST - Digital twins

nist.gov/digital-twins

Digital-twin concepts, essential elements and validation context.

IsolationForest - scikit-learn documentation

scikit-learn.org/stable/modules/generated/sklearn.ensemble.IsolationForest.html

Anomaly-model implementation reference.

Chen & Guestrin (2016) - XGBoost

arxiv.org/abs/1603.02754

Tree-boosting method used by the optional synthetic model pipeline.

TwinGuard Aero - source, model card and validation

github.com/krishnatayal1410/twinguard-aero

Inspect implementation, generic-model assumptions and reproducible checks.

References support the approach; they do not validate this prototype on real engines.